

Yibo Xu

CONTACT INFORMATION	Department of Mathematical Sciences University of South Dakota Vermillion, SD 57069 <i>Personal Webpage:</i> https://yiboxu20.github.io	<i>Office:</i> 217 Patterson Hall <i>Work:</i> (605) 658-5970 <i>Cell:</i> (775) 338-5498 <i>E-mail:</i> yiboxuopt@gmail.com or yibo.xu@usd.edu
ACADEMIC APPOINTMENTS	University of South Dakota , Vermillion, South Dakota, USA <i>Assistant Professor</i>	August 2026–Present
	University at Albany, State University of New York , Albany, New York, USA <i>Visiting Assistant Professor</i>	August 2024–August 2026
	Clemson University , Clemson, South Carolina, USA <i>Postdoctoral Fellow</i>	August 2021–August 2024
	Rensselaer Polytechnic Institute , Troy, New York, USA <i>Postdoctoral Research Associate</i>	August 2018–July 2021
	University of Nevada, Reno , Reno, Nevada, USA <i>Instructor</i>	Summer 2013; Spring 2014
EDUCATION	Clemson University , Clemson, South Carolina, USA Ph.D., Mathematical Sciences	May 2018
	• Dissertation entitled, “Convex Hulls, Relaxations, and Approximations of General Monomials and Multilinear Functions.” • Advisor: Warren Adams • GPA: 4.00/4.00	
	University of Nevada, Reno , Reno, Nevada, USA M.S., Applied Mathematics	December 2013
	• GPA: 4.00/4.00	
	Class in honor of Shiing-Shen Chern , Nankai University, Tianjin, China B.S., Mathematics and Applied Mathematics	June 2010
	• Thesis entitled, “Convexity and Optimization in Linear Spaces,” scored 95%.	
RESEARCH INTERESTS	Continuous optimization; nonlinear programming; mixed-integer programming; large-scale optimization methods for machine learning; high-precision numerical methods for nonlinear programming. Additional interests include algebra, convex analysis, computational algebraic geometry, networks, numerical analysis, numerical linear algebra, and statistics.	
PUBLICATIONS	Xu, Yangyang, Xu, Yibo , Yan, Y., Sutter-Shepard, C., Grinberg, L., and Chen, J., “Parallel and distributed asynchronous adaptive stochastic gradient methods,” <i>Mathematical Programming Computation</i> , Vol. 15, No. 3, pp. 471-508, 2023.	
	Xu, Yangyang and Xu, Yibo , “Momentum-based variance-reduced proximal stochastic gradient method for composite nonconvex stochastic optimization,” <i>Journal of Optimization Theory and Applications</i> , Vol. 196, No. 1, pp. 266-297, 2023.	

Xu, Yangyang, **Xu, Yibo**, Yan, Y., and Chen, J., “Distributed stochastic inertial-accelerated methods with delayed derivatives for nonconvex problems,” *SIAM Journal on Imaging Sciences*, Vol. 15, No. 2, pp. 550-590, 2022.

Xu, Yibo and Xu, Yangyang, “Katyusha Acceleration for Convex Finite-Sum Compositional Optimization,” *INFORMS Journal on Optimization*, Vol. 3, No. 4, pp. 418-443, 2021.

Xu, Yibo, Adams, W., and Gupte, A., “Polyhedral Analysis of Symmetric Multilinear Polynomials over Box Constraints,” arXiv:2012.06394, 2020.

Adams, W., Gupte, A., and **Xu, Yibo**, “Error bounds for monomial convexification in polynomial optimization,” *Mathematical Programming Series A*, Vol. 175, No. 1, pp. 355-393, 2019.

PAPERS IN
PREPARATION

Xu, Yibo, “Efficient and Exact Polynomial-Time Solver for the Two Trust-Region Subproblem.”

Jiang, Y., Ouyang, Y., and **Xu, Yibo**, “Exact Matrix-Vector Multiplication Complexity for Kernel Projection and Its Application on Distributed Consensus Optimization.”

Xu, Yibo, “Convex Hull Derivation for a Symmetric Multilinear Polynomial and a Symmetric Polytope.”

Xu, Yibo, Adams, W., and Gupte, A., “Error Analysis of Multilinear Terms using Linear Functions.”

UPCOMING
CONFERENCE
PRESENTATIONS

Xu, Yibo, “Momentum-Based Variance-Reduced Stochastic Gradient Method for Imaging and Large-Scale Learning,” contributed talk submitted to the 2026 Midwest Optimization Meeting, Brookings, SD, USA, October 2026.

Xu, Yibo, “Momentum-Based Variance-Reduced Stochastic Gradient Method for Imaging and Large-Scale Learning,” SIAM Conference on Imaging Science, Salt Lake City, UT, USA, November 2026.

CONFERENCE
PRESENTATIONS

Xu, Yibo, Jiang, Y., and Ouyang, Y., “Exact Matrix-Vector Multiplication Complexity for Kernel Projection and Its Application on Distributed Consensus Optimization,” INFORMS Annual Meeting, Phoenix, AZ, USA, October 2023.

Xu, Yibo and Xu, Yangyang, “Momentum-based Variance-reduced Proximal Stochastic Gradient Method for Composite Nonconvex Stochastic Optimization,” INFORMS Annual Meeting, Indianapolis, IN, USA, October 2022.

Xu, Yibo, Xu, Yangyang, Yan, Y., and Chen, J., “Distributed stochastic inertial-accelerated methods with delayed derivatives for nonconvex problems,” International Conference on Continuous Optimization, Bethlehem, PA, USA, July 2022.

Xu, Yibo and Xu, Yangyang, “Katyusha Acceleration for Convex Finite-Sum Compositional Optimization,” INFORMS Optimization Society Conference, Greenville, SC, USA, March 2022.

Xu, Yibo, Xu, Yangyang, Yan, Y., and Chen, J., “Distributed stochastic inertial-accelerated methods with delayed derivatives for nonconvex problems,” INFORMS Optimization Society Conference, Greenville, SC, USA, March 2022.

Xu, Yibo, “Deriving the Convex Hull Form of a Symmetric Multilinear Polynomial,” INFORMS Annual Meeting, Houston, TX, USA, October 2017.

Xu, Yibo, Adams, W., and Gupte, A., “Deriving Convex Hull Forms of Special Symmetric Multilinear Polynomials,” SIAM Conference on Applied Algebraic Geometry, Atlanta, GA, USA, August 2017.

Xu, Yibo, Adams, W., and Gupte, A., “On the Strength of Linear Approximations for Multilinear Monomials,” INFORMS Annual Meeting, Nashville, TN, USA, November 2016.

Xu, Yibo, Adams, W., and Gupte, A., “Error Bounds from Monomial Convexification in Polynomial Optimization,” INFORMS Annual Meeting, Nashville, TN, USA, November 2016.

SEMINAR
PRESENTATIONS

Xu, Yibo, “Stochastic First-Order Methods for Machine Learning and Imaging: Variance Reduction, Acceleration, and Distributed Computation,” Mathematical Sciences Seminar, University of South Dakota, Vermillion, SD, USA, March 2026.

Xu, Yibo, “Distributed stochastic inertial-accelerated methods with delayed derivatives for non-convex problems,” Clemson Operations Research Institute Seminar, Clemson University, Clemson, SC, USA, July 2022.

Xu, Yibo, “Solving Strongly Convex Stochastic Composition Optimization,” Dynamical Systems/RTG Seminar, Rensselaer Polytechnic Institute, Troy, NY, USA, February 2019.

Xu, Yibo, “Deriving the Convex Hull Form of a Symmetric Multilinear Polynomial,” Operations Research Seminar, Clemson University, Clemson, SC, USA, October 2017.

POSTER
PRESENTATION

Xu, Yibo, Adams, W., and Gupte, A., “Deriving Convex Hull Forms of Special Symmetric Multilinear Polynomials,” Mixed Integer Programming Workshop, Montreal, Quebec, Canada, June 2017.

PROFESSIONAL
SERVICE

Conference Session Chairing

- Co-Chair of a two-part minisymposium, SIAM Conference on Imaging Science, 2026.
- Session Chair, INFORMS Annual Meeting, 2022.
- Session Chair, International Conference on Continuous Optimization, 2022.
- Co-Chair of two sessions, INFORMS Optimization Society Conference, 2022.

Research Community Activities

- Co-organized a reading group in continuous optimization, School of Mathematical and Statistical Sciences, Clemson University, 2021–2024.

Reviewing

- *Complexity, Computational Optimization and Applications, Discrete Applied Mathematics, Discrete Dynamics in Nature and Society, IEEE Access, IEEE Transactions on Automatic Control, IEEE Transactions on Signal Processing, Journal of Global Optimization, Journal of Machine Learning Research, Journal of Scientific Computing, Mathematical Programming Series A and B, Mathematics of Operations Research, Numerical Algorithms, Numerische Mathematik.*
- *International Conference on Artificial Intelligence and Statistics (AISTATS), Integer Programming and Combinatorial Optimization (IPCO).*
- *Large Scale Optimization in Supply Chains and Smart Manufacturing: Theory and Applications*, Springer Optimization and Its Applications, 2019.
- *Recent Advances in Optimization and Modeling of Contemporary Problems*, TutORials in Operations Research, 2018.

STUDENT ADVISING Current Master’s Thesis Advising

- Bryan Jackson, University of South Dakota. Thesis research on free subgroups of orientation-preserving homeomorphisms of the real line. Advisor, 2026–Present.

Completed Doctoral Co-Advising

- Yunheng Jiang, Clemson University, 2024. Dissertation: “Efficient first-order methods for some smooth nonlinear optimization problems.” Co-Advisor.
- Yidan Guo, Clemson University, 2024. Dissertation: “First-order algorithms for convex smooth optimization problems with homogeneous linear constraints.” Co-Advisor.

TEACHING EXPERIENCE

University of South Dakota, Assistant Professor

- MATH 125 Calculus II: Fall 2026.
- Anticipated teaching includes MATH 412/512 Linear Algebra, MATH 735 Mathematical Modeling, and MATH 492/592 Topics in Applied Mathematics.

University at Albany, State University of New York, Visiting Assistant Professor

- AMAT 581 Nonparametric Statistics: Spring 2026.
- AMAT 502 Modern Computing for Mathematicians: Spring 2026.
- AMAT 592 Machine Learning: Fall 2025.
- AMAT 591 Optimization Methods and Nonlinear Programming: Spring 2025, Spring 2026.
- AMAT 500 Mathematics for Data Science: Fall 2024, Fall 2025.
- Received strong teaching evaluations, with overall instructor ratings ranging from 4.54 to 4.81 and course ratings from 4.48 to 4.63 during 2024–2026.

Clemson University, Postdoctoral Fellow and Graduate Teacher of Record

- MATH 2060 Calculus of Several Variables: Spring 2022, Spring 2023, Fall 2023, Spring 2024, Summer 2024; included large-enrollment and asynchronous online sections.
- MATH 1080 Calculus of One Variable II: Summer 2022, Fall 2022, Summer 2023; included synchronous and asynchronous online sections.
- MATH 2070 Business Calculus II: Spring 2016, Fall 2016.
- MATH 1020 Business Calculus I: Fall 2015, Fall 2017, Fall 2021.
- Teaching assistant for Calculus I, Calculus II, and Introduction to Combinatorics, 2014–2018.

University of Nevada, Reno, Instructor and Teaching Assistant

- MATH 126R Precalculus I: Summer 2013, Spring 2014; included one large-enrollment section.
- Teaching assistant for Precalculus, Calculus I, and Calculus II, 2012–2013.

HONORS AND AWARDS

Mixed Integer Programming Workshop Travel Award, \$500	2017
Outstanding MS Student, Department of Mathematical Sciences, Clemson University	2016
Institute for Mathematics and its Applications Workshop Travel Support, \$700	2016

ADDITIONAL ACADEMIC PREPARATION

Institute for Mathematics and its Applications

- New Directions Short Course: Mathematical Optimization, Minneapolis, MN, USA, August 2016.

Clemson University

- Ph.D. Preliminary examinations passed in Analysis (January 2015), Algebra (July 2015), and Operations Research (July 2015).
- Selected graduate coursework: Nonlinear Optimization Algorithms for Big Data Analysis; Advanced Linear Programming; Network Flow Programming; Discrete Optimization; Probability Theory I; Statistical Inference; Computational Algebraic Geometry; Cryptography; Matrix Analysis; Linear Analysis; Partial Differential Equations; and Finite Element Method.

University of Nevada, Reno

- Selected graduate coursework: Operations Research I and II; Graph Theory and Combinatorics; Game Theory; Cooperative Game Theory; Independent Study in Noncooperative Game Theory; Complex Function Theory; Topology I; Abstract Real Analysis I and II; Modern Algebra I and II; Numerical Analysis and Approximation I; and Methods in Applied Mathematics II.

EARLIER REFEREED PUBLICATIONS (IN CHINESE) **Xu, Yibo**, “Other Solutions for the 2nd Problem in the Second Round of National Senior High School’s Mathematical Competition of China in 2005,” *High-School Mathematics*, Vol. 26, No. 8, pp. 19-21, 2007.

Xu, Yibo and Yu, S., “Proof of a set of inequalities,” *Bulletin of Mathematics (Wuhan Mathematical Society)*, No. 18, pp. 46-47, 2004.

EARLIER LEADERSHIP AND OUTREACH **Siguo Wargame Association, Nankai University**, Tianjin, China
Chairman **September 2009–June 2010**

Nankai Bulletin Board System, Nankai University, Tianjin, China
Moderator of Board “Mathematics” and Board “Siguo Wargame” **September 2008–June 2010**

Xutuan Middle School, Xutuan Town, Mengcheng County, Anhui Province, China
Volunteer Math Teacher **June 2007–July 2007**

COMPUTER SKILLS Python, C++, MATLAB; AMPL, Mathematica, Maple, CVX, MPLAPACK, PORTA; and L^AT_EX.

Updated September 14, 2026